

Adaptive Deformable Features for Leaf Vein Segmentation

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Abstract. Extracting traits from plant leaves (e.g. vein architecture) is important for genotype and phenotype studies. However, leaf vein segmentation remains challenging due to the randomness, structural complexity, and high morphological diversity across cultivars, where veins are often difficult to distinguish even from the surrounding lamina. To address this problem, we propose AdaptNet, a decoder architecture built upon a novel adaptive feature extraction module that captures global context, fine local details, as well as subpixel vein structures through both deformable and atrous convolutional layers. Unlike conventional decoders with consistent layer widths at every stage, our decoder dynamically adapts layer widths via neural architecture search, enabling dataset specific optimization of semantic and localization features across decoding stages. We evaluate AdaptNet on three leaf vein segmentation benchmarks, achieving top performance with an mIoU of 82.46%, while using nearly half the number of parameters in baselines using a conventional DenseNet161-based U-Net. Notably, even when using a relatively old DenseNet161 encoder, AdaptNet surpasses modern transformer-based segmentation models, including DPT with DINOv2 and DINOv3 encoders, as well as, recent ConvNet architectures such as ConvNeXt-based UPerNet. Our results demonstrate the effectiveness of our adaptive decoder design for fine-grained vein segmentation across diverse plant cultivars.

Keywords: Neural architecture search, Semantic segmentation, Fine-grained segmentation

1 Introduction

Studying leaf vein architecture is essential because it plays a critical role in leaf function, including water transport, nutrient distribution, drought resistance, and photosynthesis. The plant vascular system directly influences resource distribution and, consequently, plant growth, yield, and stress tolerance [24]. For example, vein diameter affects transport capacity [27] and mechanical support [18]. In specific crops such as soybeans, studies of vein traits offer potential for breeding programs that target drought resilience and photosynthetic efficiency under climate stress [17]. Moreover, vein traits can serve as morphological markers for genetic mapping and plant classification [20]. Therefore, improving image-based methods for vein segmentation is a prerequisite for reliable and robust

trait extraction, thereby accelerating plant genetics-driven improvements and adaptations.

Semantic segmentation is one of the main tasks in computer vision, supporting many applications. Although related to image classification, it differs by generating a dense prediction, assigning a class label to each pixel rather than a single label to the entire image. This makes labeling for segmentation far more labor-intensive than for classification, which is why classification datasets such as ImageNet [8], with 1000 classes, are much larger and more diverse than segmentation benchmarks such as ADE20K [34] and Cityscapes [6], which contain only 150 and 30 classes, respectively. Fortunately, features learned for classification are largely transferable to segmentation [16]. A common approach is therefore to use ConvNets or transformers pre-trained on large classification datasets as encoders in encoder-decoder segmentation models [13, 33], leveraging their ability to capture general-purpose visual features without training from scratch.

The encoder, often referred to as the backbone, extracts features at multiple spatial resolutions, which the decoder then merges to produce the final prediction mask. Beyond supervised classification encoders, self-supervised encoders such as DINOv2 [22] have also proven effective as backbones for segmentation [32]. However, even though advances in ConvNet and transformer architectures have driven progress in semantic segmentation [9, 14], recent work argues that relying on outdated encoders hinders further gains [32]. In particular, Yang et al. [32] show that simply upgrading the encoder, sometimes with even fewer parameters, can yield greater improvements than complex method redesigns, and they demonstrate this by building their state-of-the-art semi-supervised segmentation method, UniMatchV2, on the DINOv2 backbone. Notably, however, most of these advances focus on the encoder side, while the potential impact of the decoder on segmentation performance remains largely overlooked [14, 32].

Despite advances in deep learning and segmentation models in particular, most architectures are optimized for segmenting coarse, blob-like structures. Relatively little research has focused on the challenging task of segmenting fine or extremely thin objects, such as leaf veins. Vein segmentation demands that the model capture fine-grained features at the pixel level to segment individual veins, while simultaneously understanding the larger-scale structure of the lamina and the hierarchical relationships among vein orders (primary, secondary, tertiary, and higher-order veins). The margin for error is exceptionally small, a prediction off by even one or two pixels can cause a tertiary vein to be misclassified as a lamina. Achieving accurate and reliable vein segmentation is, therefore, a prerequisite for comprehensive trait extraction from plant leaves. Only then can veins be classified by order, and their spatial relationships (e.g., insertion angles between primary and secondary veins) be quantified for use in genetic and physiological studies of plants.

Another limitation of typical segmentation models in this context is that ConvNet decoders generally use convolutional layers with a fixed receptive field at each stage. A fixed receptive field is not ideal for capturing semantic information that spans vastly different spatial scales within the same image. This

is a challenge that is particularly acute in vein segmentation, where structures range from thick primary veins to tertiary veins only one or two pixels wide, and where global context is needed to distinguish among the hierarchy of vein orders and the surrounding lamina. Equipping the decoder with the ability to both expand its receptive field for broader context and capture subpixel-level details for thin structures can therefore lead to more spatially precise and semantically consistent segmentation.

This research focuses on designing a decoder based on a novel adaptive module. The proposed module combines deformable and atrous convolution layers, bringing together complementary feature extraction components that enhance the model’s ability to recover fine details and capture distributed spatial patterns. To determine the optimal width for each layer and maximize performance on a given dataset, we conduct a Neural Architecture Search (NAS). Varying layer widths expands or shrinks the feature space, enabling the model to learn richer representations while discarding irrelevant ones. However, increasing the width beyond a certain point can degrade performance, as the feature space grows too large relative to the dataset size, leading to insufficient training signal. We therefore design our model to be adaptable and to explore a range of possible configurations, allowing us to identify the most effective channel widths for each layer and enhance the model’s capacity to capture relevant features for a given dataset. Our approach, even when using a relatively old encoder as the backbone, outperforms state-of-the-art models.

The key contributions of our work can be summarized as follows:

1. We propose a new module that combines deformable and atrous convolution layers for semantic segmentation. Using this module as a building block, we present a novel decoder capable of extracting richer features from very fine objects.
2. We propose the use of NAS to adapt component widths in the decoder. Thanks to this adaptable design, the decoder offers the flexibility to adjust its structure and size to extract the local and global features most relevant to a given dataset.
3. We evaluate our model against state-of-the-art transformer and ConvNet-based models. We demonstrate that even when using a 10-year-old encoder, our model outperforms architectures such as DPT [23] equipped with modern encoders like DINOv2 [22] (e.g., as implemented in UniMatch V2 [32]).

The source code and datasets used in this research will be made publicly available at a URL to be disclosed upon acceptance of the paper (double-blind rules).

2 Proposed Method

2.1 Encoder

Like other studies that use pre-trained models to accelerate training, our architecture uses DenseNet161 [12] as the backbone. Our pipeline extracts multi-scale

feature maps from various layers within this backbone (F_{0-4}) and feeds them into the decoder, as shown in Figure 1. Each feature map carries different levels of spatial detail and semantic information. Specifically, deeper layers capture more semantic features with lower spatial precision, while shallower layers preserve finer spatial details with less semantic content. The initial feature map (F_0) retains most of the spatial detail, and subsequent layers progressively abstract and condense this information. As a result, F_0 has the largest spatial resolution, while F_4 has one-sixteenth the resolution of F_0 .

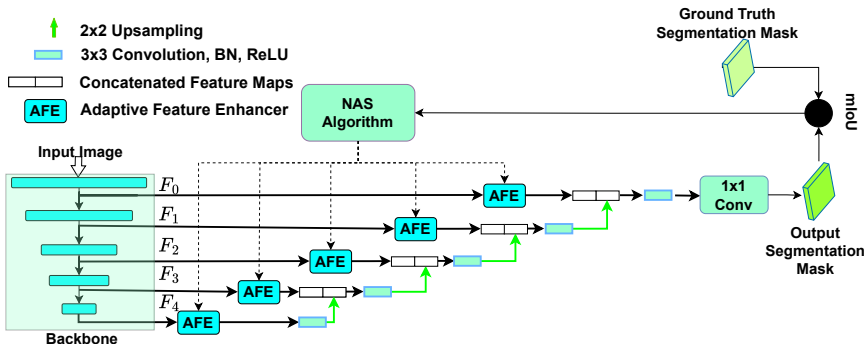


Fig. 1: Schematic diagram of the complete model with both classification and segmentation heads

2.2 Adaptive Feature Extraction Module

When segmenting objects in ultra or fine-grained datasets, such as leaf veins and other minute objects in nature, the diminished representation caused by repeated downsampling may make leaf veins indistinguishable from other veins in the hierarchy or even from the lamina. To address this problem, we propose a novel adaptive feature extraction module that operates over the multi-scale feature maps, as shown in Figures 1 and 2. The goal is to capture the minute features of leaf veins while preserving their broader spatial relationships. The module in Figure 2 is shared across F_0 to F_4 , as shown in Figure 1. The module integrates two sizes of deformable convolution layers and atrous convolution layers [7], adopting a design similar to the InceptionNet module presented in [26].

The deformable convolution layer has an adaptive receptive field that captures spatial details across different scales, offering capabilities similar to global attention, such as capturing long-range dependencies, while also capturing sub-pixel details through the adaptive spatial aggregation. By learning offset parameters, deformable convolution layers can handle scale variations as well as complex spatial arrangements. Figure 3 illustrates the receptive fields of three

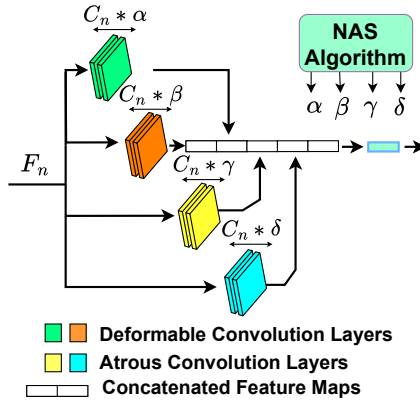


Fig. 2: Adaptive Feature Extraction Module

different convolutional layers. The first is self-attention, which aggregates information from a broad area around the query pixel P_0 , providing wide, global context. The second is a standard 3×3 convolution, which focuses on a small, local neighborhood around P_0 , capturing only adjacent information. The third is deformable convolution, which can capture features from distant pixels or sub-pixels, allowing for flexible and efficient aggregation by adjusting its receptive field to relevant features in the feature maps extracted from the backbone.

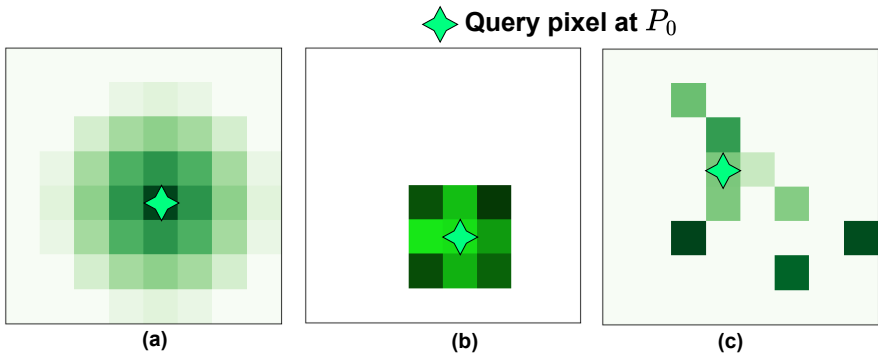


Fig. 3: Comparison between receptive fields of different convolution layers: (a) global aggregation with multi-head self-attention, (b) the standard 3×3 convolution, and (c) a deformable convolution that combines properties of multi-head self-attention with enhanced efficiency.

However, deformable convolution is much more computation and memory efficient than global attention [29]. The deformable convolution layer can be described by the following equation:

$$y(\mathbf{p}_0) = \sum_{\mathbf{p}_n \in \mathcal{R}} \mathbf{w}(\mathbf{p}_n) \cdot \mathbf{x}(\mathbf{p}_0 + \mathbf{p}_n + \Delta\mathbf{p}_n) \quad (1)$$

where the set $\mathcal{R} = \{(-1, -1), (-1, 0), (-1, 1), (0, -1), (0, 0), (0, 1), (1, -1), (1, 0), (1, 1)\}$ defines the regular grid offsets for a 3×3 kernel, $\mathbf{w}(\mathbf{p}_n)$ is the weight at position $\mathbf{p}_n$, and $\Delta\mathbf{p}_n$ is a learned offset produced by an accompanying convolution branch. Since the predicted offset $\Delta\mathbf{p}_n$ is typically fractional, the sampling position $\mathbf{p} = \mathbf{p}_0 + \mathbf{p}_n + \Delta\mathbf{p}_n$ falls at a subpixel location. Therefore, the feature value $\mathbf{x}(\mathbf{p})$ is obtained via bilinear interpolation:

$$\mathbf{x}(\mathbf{p}) = \sum_{\mathbf{q}} G(\mathbf{q}, \mathbf{p}) \cdot \mathbf{x}(\mathbf{q}) \quad (2)$$

where $\mathbf{q}$ enumerates all integer spatial locations in the input feature map $\mathbf{x}$, and $G(\mathbf{q}, \mathbf{p})$ is the bilinear interpolation kernel.

Meanwhile, the atrous convolution layers encode multi-scale contextual information by using different preset dilation rates, each providing a different field of view. They enable the model to efficiently capture details at various scales without increasing computational cost or model size [4]. This is achieved by inserting zeros between the filter weights, effectively dilating the kernel and creating dependencies between features at multiple scales. Atrous convolution can be expressed as:

$$y(\mathbf{p}_0) = \sum_{\mathbf{p}_n \in \mathcal{R}} \mathbf{w}(\mathbf{p}_n) \cdot \mathbf{x}(\mathbf{p}_0 + r \cdot \mathbf{p}_n) \quad (3)$$

where $y(\mathbf{p}_0)$ is the output at position $\mathbf{p}_0$, $\mathcal{R}$ represents the set of relative positions in the filter, $\mathbf{w}(\mathbf{p}_n)$ denotes the filter weights, $\mathbf{x}(\cdot)$ is the input feature map, and r is the dilation rate that determines the spacing between filter elements. This dilation allows the filter to capture a larger context around $\mathbf{p}_0$.

The feature extraction module integrates two deformable convolution layers, one with a 3×3 kernel and the other with a 5×5 kernel. The width of these layers is adaptive, meaning the number of output channels is adjustable. This adaptive width is determined by the NAS algorithm through parameters α and β , which control the number of output channels as illustrated in Figure 2. Additionally, we employ two atrous convolution layers with a 3×3 kernel size but with different dilation rates of two and three, to capture contextual information at multiple scales. As with the deformable convolution layers, the number of output channels from the atrous convolution layers is governed by two parameters γ and δ , also determined by the NAS algorithm, as shown in Figure 2. This enables the model to tune the number of output channels from both the deformable and atrous convolution layers, allowing the network to adjust the capacity and complexity of its layers to suit different tasks and datasets and capture the necessary semantic and spatial detail in each feature representation.

2.3 Neural Architecture Search

The adaptive feature extraction layer at stage n can be represented as a function $Z_n = \mathcal{A}_n(F_n)$, where Z_n denotes the output tensor and F_n represents the input feature map with shape $\langle H_n, W_n, C_n \rangle$. Here, C_n is the number of channels, and H_n and W_n are the spatial dimensions. The output tensor from any of the five stages of the complete model in Figure 1 can be described as follows:

$$Y_n(Y_{n+1}, Z_n) = \begin{cases} \mathcal{F}^{1 \times 1}(\mathcal{F}_n^{3 \times 3}([\mathcal{F}_{\text{up}}(Y_{n+1}) \parallel Z_n])) & n = 0 \\ \mathcal{F}_n^{3 \times 3}([\mathcal{F}_{\text{up}}(Y_{n+1}) \parallel Z_n]) & 0 < n < 4 \\ \mathcal{F}_n^{3 \times 3}(Z_n) & n = 4 \end{cases} \quad (4)$$

where $\mathcal{F}_n^{3 \times 3}$ and $\mathcal{F}^{1 \times 1}$ are 3×3 and 1×1 convolution layers respectively, $\mathcal{F}_{\text{up}}$ is a 2×2 up-convolution, and $[\parallel]$ denotes the concatenation operation along the channel dimension. As shown in Figures 2 and 1, the number of output channels is adaptive, since the importance of the deformable and atrous convolution operations may vary across different stages.

The dimensions of Z_n are $H_n \times W_n \times (C_n \cdot \Psi)$, where $\Psi = \alpha + \beta + \gamma + \delta$ and $(\alpha, \beta, \gamma, \delta)$ are fractional parameters controlling the number of output channels from each convolution type, as illustrated in Figure 2. The complete decoder can be represented as:

$$\mathcal{M} = \bigodot_{n=0 \dots 4} Y_n \left(Y_{n+1}, Z_n^{\langle H_n, W_n, C_n \cdot \Psi \rangle} \right) \quad (5)$$

The objective of NAS is to maximize the mIoU, as shown in the following:

$$\begin{aligned} & \max_{\alpha, \beta, \gamma, \delta} \text{mIoU}(\mathcal{M}(\alpha, \beta, \gamma, \delta)) \\ \text{s.t. } & \mathcal{M} = \bigodot_{n=0 \dots 4} Y_n \left(Y_{n+1}, Z_n^{\langle H_n, W_n, C_n \cdot \Psi \rangle} \right) \end{aligned} \quad (6)$$

To this end, we employ Optuna for Neural Architecture Search [1], utilizing its Tree-structured Parzen Estimator (TPE) sampler [2] to determine the number of output channels from each of the deformable and atrous convolution layers for each dataset.

TPE models the probability distribution of hyperparameters x with respect to the objective function mIoU. It classifies the observed hyperparameters into two groups based on a threshold γ_{tpe} , representing a quantile of the observed objective values. The two distributions are: $l(x)$ for observations above γ_{tpe} (associated with higher objective values) and $g(x)$ for observations below γ_{tpe} (associated with lower objective values).

Hyperparameters are selected by maximizing the Expected Improvement (EI), which is derived from the ratio of these two distributions:

$$EI(x) = \frac{l(x)}{g(x)}$$

where x represents the hyperparameters, $l(x) = P(x \mid y > \gamma_{\text{tpe}})$ models the distribution of x leading to outcomes better than the threshold γ_{tpe} , and $g(x) = P(x \mid y \leq \gamma_{\text{tpe}})$ models the distribution of x leading to outcomes worse than the threshold.

The NAS algorithm determines α , β , γ , and δ , which are fractional parameters that maximize performance while maintaining efficiency. These fractions control the number of output channels of the deformable and atrous convolution layers. For example, the number of output channels of the 3×3 deformable convolution is computed as $C_0 \cdot \alpha$, where C_0 is the number of channels of the input feature map F_0 and α is determined by the NAS algorithm, as shown in Figure 2.

3 Results

To evaluate our proposed approach thoroughly, we trained and tested multiple models on three benchmark datasets: IHS [3], SoyCultivar200 (SC200) [28], and our developed dataset Soy25, which contains 110 training images and 31 testing images. All segmentation models were trained and evaluated under the same experimental setup and data splits. We compared our model against state-of-the-art ConvNet- and Transformer-based segmentation methods. Specifically, the Transformer-based approaches include UPerNet with Swin backbones [14] and SegFormer with MiT backbones [31]. We also initialized these models with different weight configurations, including random initialization and pre-trained weights from ADE20K [34]. Moreover, we evaluated DPT [23] with a ViT backbone [9], initialized using self-supervised weights from DINOv2 [22] and DINOv3 [25].

Mechanism	IHS	SC200	Soy25	# (M) Paras.	IoU
(Our Baseline) Dense-UNet	78.50	81.55	83.99	155	81.35
Simple Channel Attention (NAFNet) [5]	77.70	80.83	82.02	165	80.18
Triple Attention [19]	78.24	81.48	83.61	155	81.11
Channel Attention (CBAM) [30]	78.41	81.57	83.57	156	81.18
Spatial Attention [21]	78.30	81.98	83.97	175	81.42
Strip Pooling [11]	79.38	82.53	84.09	195	82.00
AdaptNet (Ours)	79.79	82.98	84.61	74	82.46

Table 1: Comparison of different attention and pooling mechanisms integrated into the Dense-UNet baseline across three leaf vein segmentation datasets, reporting IoU and number of parameters, with AdaptNet achieving the best overall performance and efficiency

Meanwhile, the ConvNet-based methods include SegNeXt [10], UPerNet with ConvNeXt backbones [15], and DeepLabV3+ with a ResNet-101 encoder. These models are also initialized with different weight configurations, including pre-trained weights from ADE20K [34], Cityscapes, STARE, and DRIVE, as well

Method	Backbone	Weights	IHS	SC200	Soy25	# (M) Paras.	IoU
DPT	ViT-Base	-	70.55	68.23	77.19	98	71.99
DPT	ViT-Base	DINOV2	80.46	80.59	83.55	98	81.53
DPT	ViT-Base	DINOV3	80.06	<u>80.65</u>	82.22	123	80.98
AdaptNet (Ours)	DenseNet161	ImageNet	79.79	82.98	84.61	74	82.46

Table 2: Comparison of AdaptNet with DPT variants using different ViT backbones and pretrained weights (DINOV2 and DINOV3), reporting IoU and number of parameters, with AdaptNet achieving competitive accuracy with fewer parameters.

as random initialization. By evaluating a range of ConvNet- and Transformer-based segmentation methods, backbones, and pre-training strategies, we provide a thorough performance comparison across diverse segmentation architectures.

Table 1 compares different attention and pooling mechanisms integrated into the Dense-UNet baseline. While all mechanisms provide moderate improvements, AdaptNet consistently achieves the highest IoU across all three datasets (IHS, SC200, and Soy25) with significantly fewer parameters (74M vs. 155–195M). This demonstrates that the proposed adaptive decoder design is more effective and parameter-efficient than conventional channel, spatial, or pooling-based attention modules.

Table 2 evaluates AdaptNet against DPT variants employing different ViT backbones and pre-trained weights (DINOV2 and DINOV3). Although larger DPT models can reach competitive IoU, they do so at a higher computational cost, using up to 123M parameters compared to AdaptNet’s 74M. AdaptNet matches or surpasses most DPT configurations on SoyCultivar200 and Soy25, demonstrating that a lightweight CNN backbone combined with the proposed decoder can outperform large Transformer-based models, contrary to the findings of [32].

Table 3 extends the comparison to a broader set of Transformer- and ConvNet-based segmentation models. AdaptNet outperforms all competing methods on each dataset and in overall IoU, while maintaining a relatively small number of parameters. These results confirm that AdaptNet achieves a superior balance between accuracy and efficiency, making it particularly well suited for fine-grained leaf vein segmentation.

4 Conclusion

Leaf vein segmentation is essential for plant genetic analysis and phenotyping, yet it remains challenging due to the hierarchical complexity of vein structures and substantial variability across cultivars. In this paper, we introduced AdaptNet, a decoder architecture based on a novel module combining deformable and atrous convolutions to jointly capture global context, fine local features, and subpixel vein details. Unlike conventional fixed-width decoders, our design utilizes neural architecture search to automatically adapt layer widths across decoding stages, balancing semantic representation and localization accuracy.

	Method	Backbone	Weights	IHS	SC200	Soy25	# (M) Paras.	IoU
Transformer	UPerNet	Swin-S	ADE20K	77.85	71.15	71.66	81	73.55
	UPerNet	Swin-B	ADE20K	77.83	70.86	72.29	121	73.66
	SegFormer	MiT-B5	-	71.54	55.62	52.01	82	59.72
	SegFormer	MiT-B4	ADE20K	78.27	66.56	72.99	61	72.61
	SegFormer	MiT-B5	ADE20K	78.13	68.91	74.29	82	73.78
ConvNet	SegNext	MSCAN-S	ADE20K	76.97	41.27	51.54	14	56.60
	SegNext	MSCAN-L	ADE20K	76.25	42.43	48.47	<u>49</u>	55.72
	UPerNet	ConvNeXt-B	ADE20K	77.09	70.80	70.54	122	72.81
	UPerNet	ConvNeXt-L	ADE20K	76.94	70.35	70.77	235	72.69
	UPerNet	ConvNeXt-XL	ADE20K	77.59	69.46	69.72	391	72.26
	DeepLabV3+	ResNet-101	-	75.48	74.24	71.76	63	73.83
	DeepLabV3+	ResNet-101	Cityscapes	75.92	77.65	76.85	63	76.81
	DeepLabV3+	ResNet-101	ADE20k	76.17	78.94	77.65	63	77.59
	AdaptNet (Ours)	DenseNet161	ImageNet	79.79	82.98	84.61	74	82.46

Table 3: Comparison of AdaptNet with Transformer- and ConvNet-based segmentation models, reporting IoU and number of parameters, where AdaptNet achieves the best overall performance with a lightweight backbone.

Experiments on three benchmark datasets show that AdaptNet achieves state-of-the-art segmentation performance with significantly fewer parameters, outperforming modern Transformer-based approaches such as DPT with DINOv2 and DINOv3, as well as strong ConvNet baselines including ConvNeXt-based UPerNet. This work demonstrates the potential of adaptive decoder architectures for accurate and robust ultra-fine-grained object extraction.

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